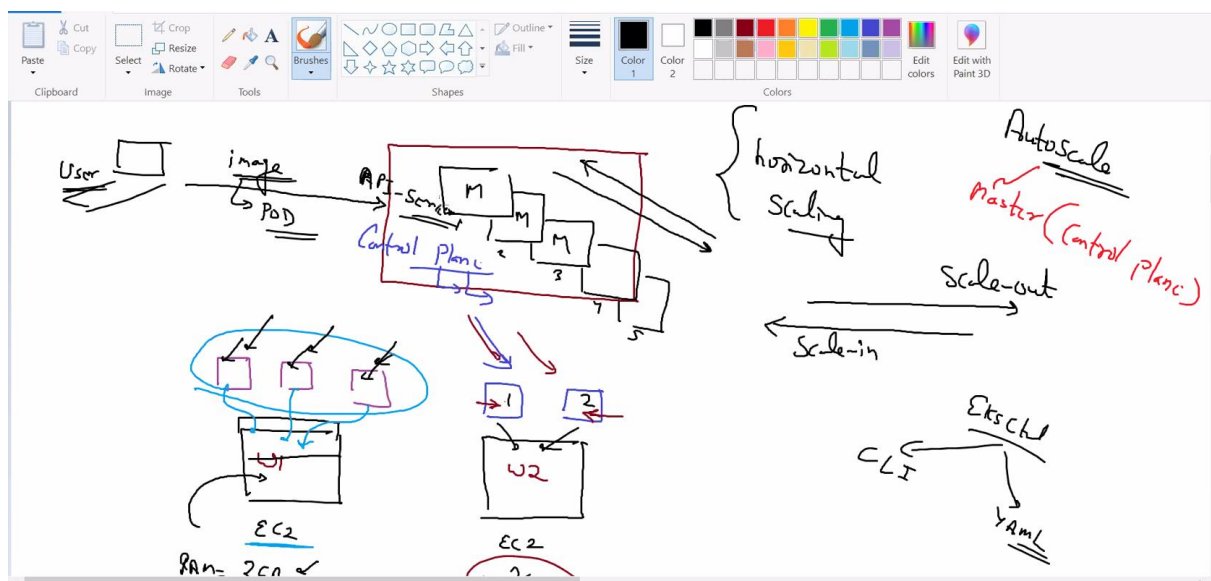




AWS Session

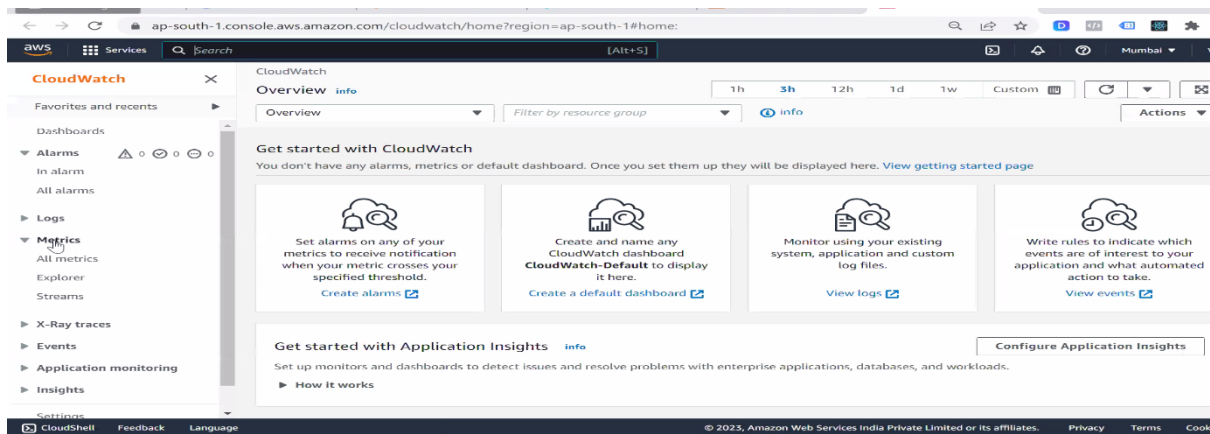
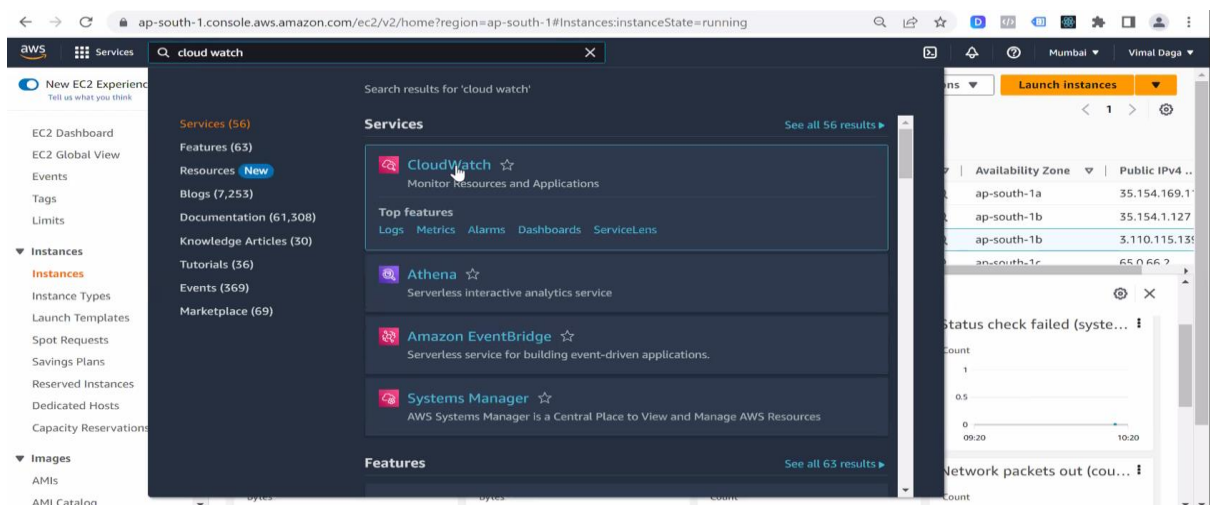
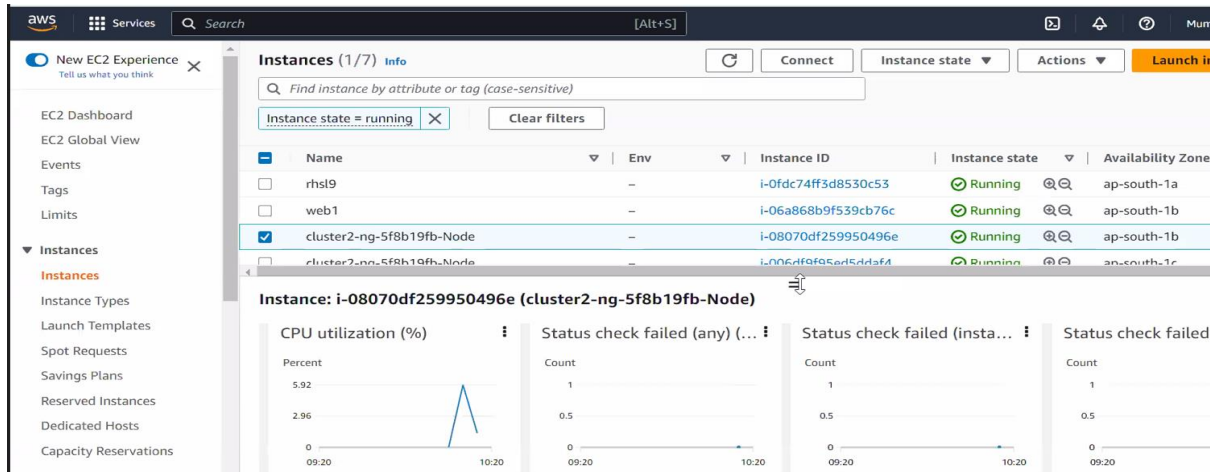
Summary - 2-4-2023

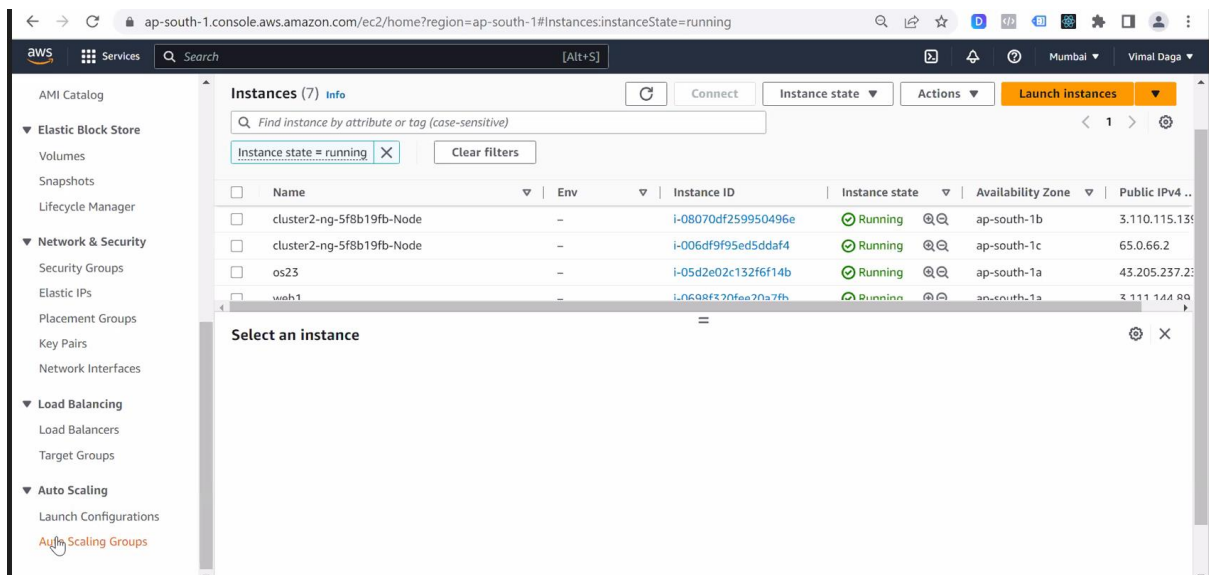
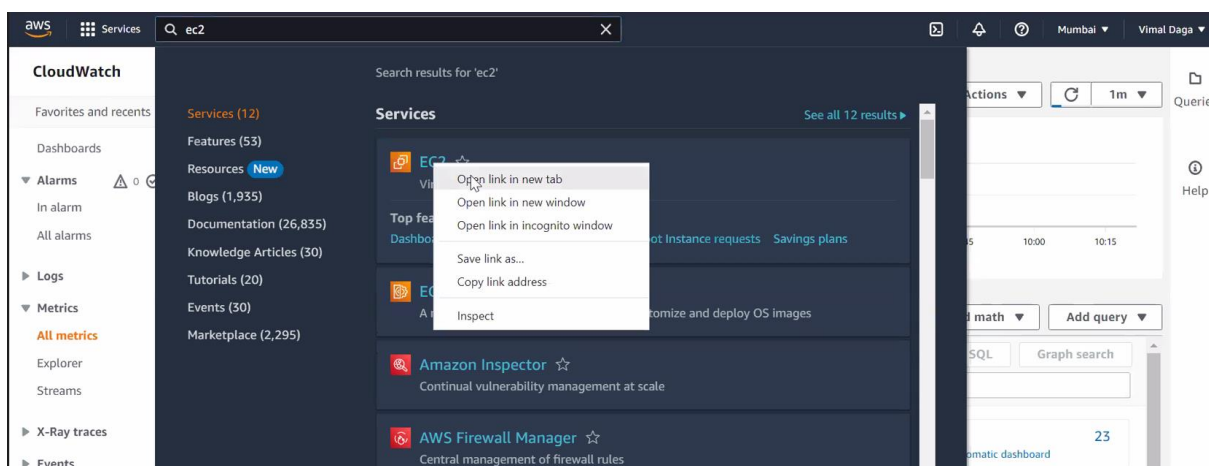
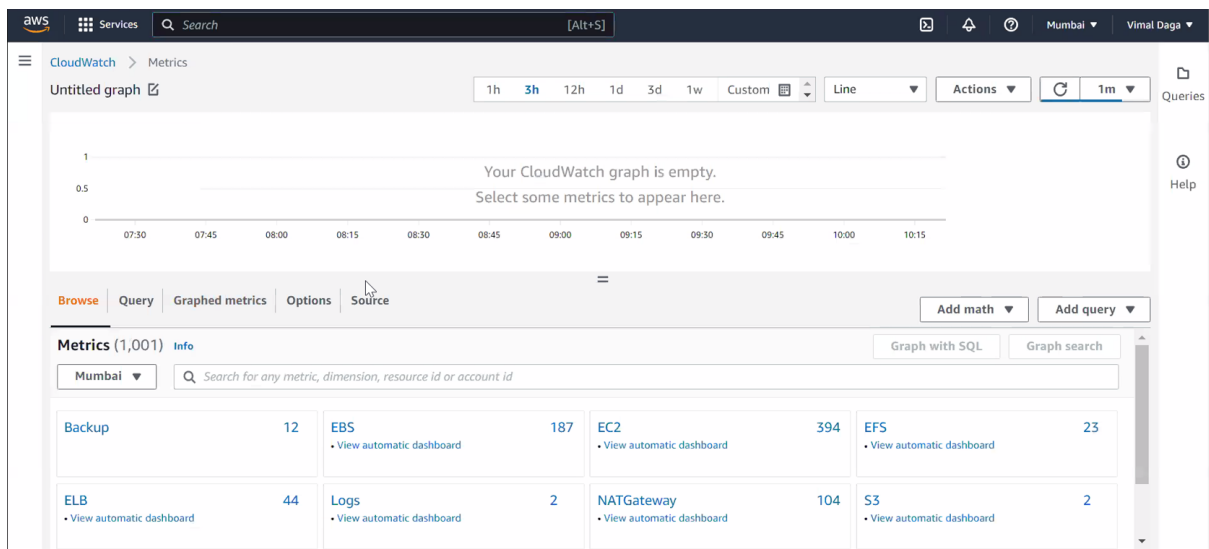
- EKS helps to integrate with other services like EC2, EBS, EFS.
- To implement EKS we use either Graphical Method or eksctl tool.
- AUTOSCALING:
 1. Cluster Node
 2. Pod HPA
 3. Eksctl – yaml
- Master is more like serverless in EKS. Entire management of master will be run by AWS internally.



- Master is more like serverless in EKS. Entire management of master will be run in AWS internally.
- Adding computers is known as **horizontal scaling**. If we increase the computers than that is known as **Scale out**.
- Decreasing the resources is called **Scale in**.
- If load increase or decrease, AWS will automatically scale that is known as **Autoscaling**.
- We have scheduler in master (control plane) and they will schedule automatically.

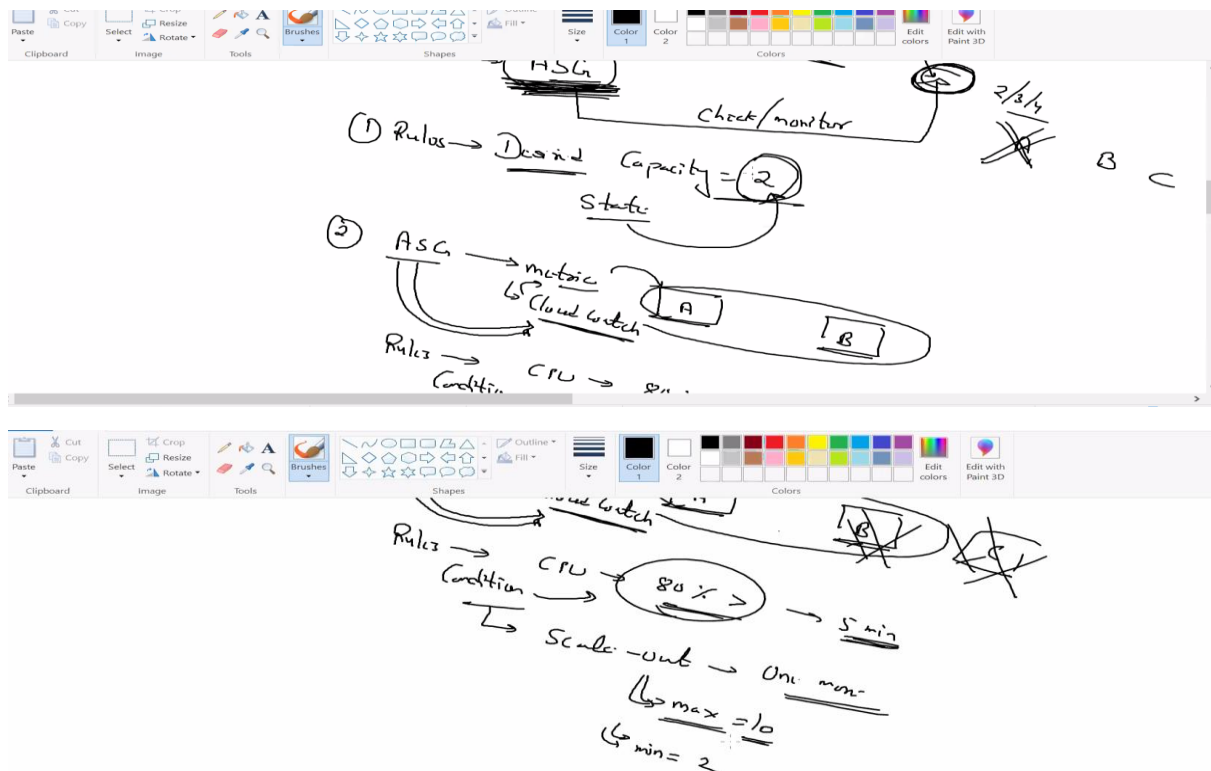
- If we keep on checking how much RAM, memory, how much I/O then that is known as **Metrics**.
- Admin keeps on monitoring the node, they launch when requirement comes up. This concept is known as Scaling.
- When we have to manual scale, we have to check manually information about nodes. We can also use some tools to monitor like prometheus, Grafana(information in the form of graph).
- Cloudwatch checks metrics on real time.



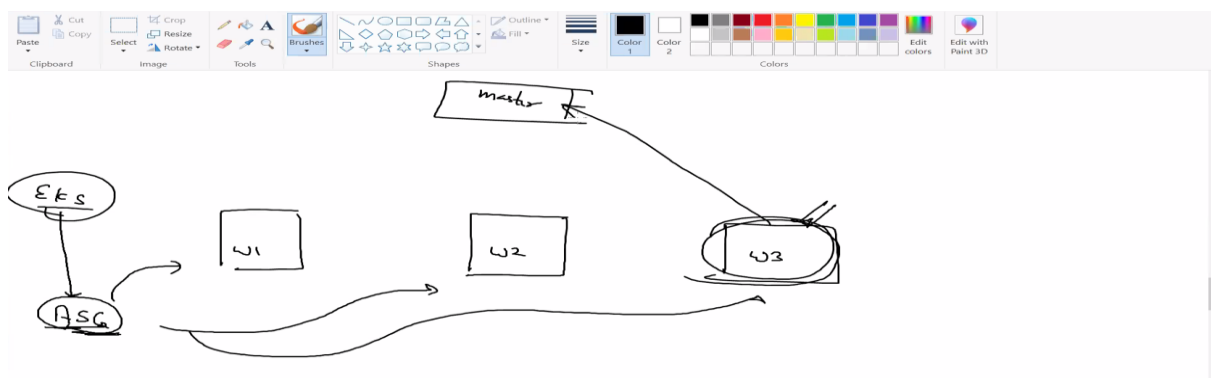


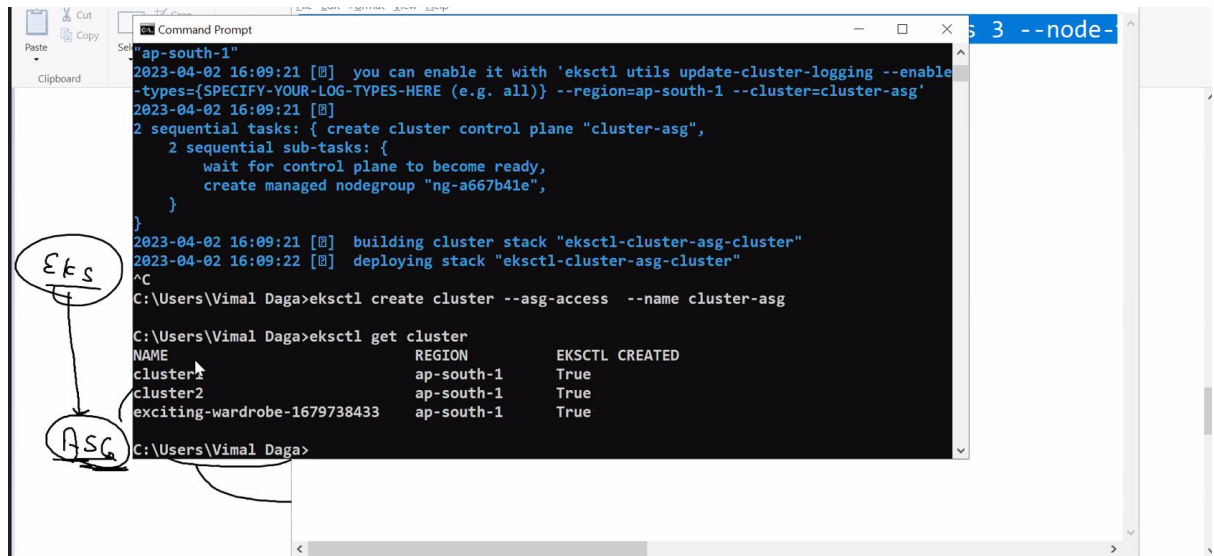
- How to do autoscaling of worker node.
This concept is not managed by k8s, it is not the role of k8s because k8s is running inside it.

- To autoscale we have a service in AWS called ASG – Auto Scaling groups, which helps us to autoscale instance.
- If we want to launch ec2 we have two ways:
 1. Manually launch
 2. ASG
- In ASG, it launch instance, keeps on monitoring the instance and whenever load increase it autoscale. If any of the instance goes down ASG will launch new instance.
- ASG keeps on monitoring the nodes metrics, behind the seen it uses metrics.



- While launching cluster, we can launch worker node with the help of ASG.





```

C:\Users\Vimal Daga>eksctl create cluster --asg-access --name cluster-asg

C:\Users\Vimal Daga>eksctl get cluster
NAME      REGION    EKSCTL CREATED
cluster1  ap-south-1  True
cluster2  ap-south-1  True
exciting-wardrobe-1679738433  ap-south-1  True

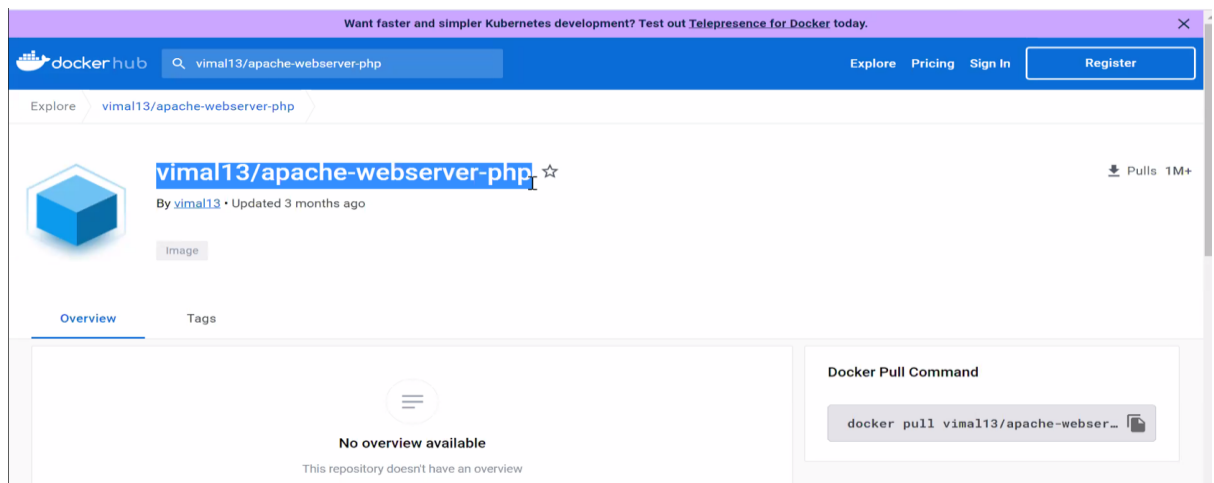
```

Handwritten notes: **EKS** (circled) and **ASG** (circled) with an arrow pointing from EKS to ASG.

```

C:\Users\Vimal Daga>eksctl create cluster --asg-access --name my-cluster-asg --asg-access --nodes-max 10 --nodes-min 2 --nodes 3 --node-type t2.micro --nodegroup-name ngA --ssh-access --enable-ssm --version 1.25 --region us-east-1
2023-04-02 16:11:27 [!] SSM is now enabled by default; `ssh.enableSSM` is deprecated and will be removed in a future release

```



Want faster and simpler Kubernetes development? Test out Telepresence for Docker today.

dockerhub vimal13/apache-webserver-php Explore Pricing Sign In Register

Explore vimal13/apache-webserver-php

vimal13/apache-webserver-php ☆ Pulls 1M+

By vimal13 • Updated 3 months ago

Image

Overview Tags

No overview available
This repository doesn't have an overview

Docker Pull Command

```
docker pull vimal13/apache-webserver-php
```

- Using apache-webserver image-php to create deployment

```

C:\Users\Vimal Daga>kubect1 create deployment myd --image=vimal13/apache-webserver-php
deployment.apps/myd created

C:\Users\Vimal Daga>

```

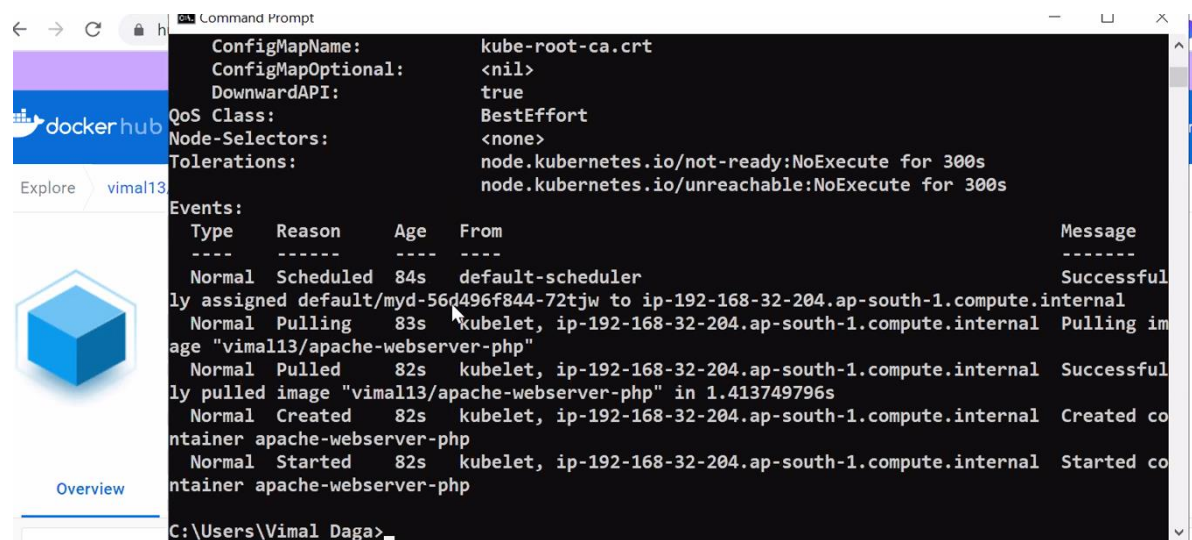
```
C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myd-56d496f844-tttmr               1/1     Running   0           27s

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubectl scale deployment myd --replicas=5
deployment.apps/myd scaled
```

```
C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myd-56d496f844-72tjw               1/1     Running   0           10s
myd-56d496f844-j6ldp               1/1     Running   0           10s
myd-56d496f844-rqgb9               1/1     Running   0           10s
myd-56d496f844-tttmr               1/1     Running   0           94s
myd-56d496f844-wqnx               1/1     Running   0           10s

C:\Users\Vimal Daga>kubectl describe pods myd-56d496f844-72tjw
```



The image shows a Docker Hub page on the left and a Windows Command Prompt on the right. The Docker Hub page displays the repository 'vimal13/apache-webserver-php' with an 'Overview' tab selected. The Command Prompt shows the output of the command 'kubectl describe pods myd-56d496f844-72tjw'.

```

ConfigMapName: kube-root-ca.crt
ConfigMapOptional: <nil>
DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute for 300s
              node.kubernetes.io/unreachable:NoExecute for 300s

Events:
  Type     Reason      Age   From
  ----     -
  Normal   Scheduled   84s   default-scheduler
  ly assigned default/myd-56d496f844-72tjw to ip-192-168-32-204.ap-south-1.compute.internal
  Normal   Pulling     83s   kubelet, ip-192-168-32-204.ap-south-1.compute.internal
  age "vimal13/apache-webserver-php"
  Normal   Pulled      82s   kubelet, ip-192-168-32-204.ap-south-1.compute.internal
  ly pulled image "vimal13/apache-webserver-php" in 1.413749796s
  Normal   Created     82s   kubelet, ip-192-168-32-204.ap-south-1.compute.internal
  Created container apache-webserver-php
  Normal   Started     82s   kubelet, ip-192-168-32-204.ap-south-1.compute.internal
  Started container apache-webserver-php
  
```

```
C:\Users\Vimal Daga>kubectl describe pods myd-56d496f844-72tjw
Name: myd-56d496f844-72tjw
Namespace: default
Priority: 0
Node: ip-192-168-32-204.ap-south-1.compute.internal/192.168.32.204
Start Time: Sun, 02 Apr 2023 16:16:40 +0530
Labels: app=myd
        pod-template-hash=56d496f844
Annotations: kubernetes.io/psp: eks.privileged
Status: Running
IP: 192.168.63.153
IPs:
  IP: 192.168.63.153
Controlled By: ReplicaSet/myd-56d496f844
Containers:
  apache-webserver-php:
    Container ID: containerd://050b32860fdadb7ad07ffd56fccd0b4a7bda45d52812a15aae2928f3c602
```

```
C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myd-56d496f844-72tjw               1/1     Running   0           97s
myd-56d496f844-j6ldp               1/1     Running   0           97s
myd-56d496f844-rqgb9               1/1     Running   0           97s
myd-56d496f844-tttmr               1/1     Running   0           3m1s
myd-56d496f844-wqnx               1/1     Running   0           97s

C:\Users\Vimal Daga>kubectl scale deployment myd --replicas=6
deployment.apps/myd scaled

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
myd-56d496f844-72tjw               1/1     Running             0           2m22s
myd-56d496f844-j6ldp               1/1     Running             0           2m22s
myd-56d496f844-q4n5w               0/1     ContainerCreating   0           3s
myd-56d496f844-rqgb9               1/1     Running             0           2m22s
myd-56d496f844-tttmr               1/1     Running             0           3m46s
myd-56d496f844-wqnx               1/1     Running             0           2m22s

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubectl scale deployment myd --replicas=4
deployment.apps/myd scaled

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myd-56d496f844-j6ldp               1/1     Running   0           3m30s
myd-56d496f844-q4n5w               1/1     Running   0           71s
myd-56d496f844-tttmr               1/1     Running   0           4m54s
myd-56d496f844-wqnx               1/1     Running   0           3m30s

C:\Users\Vimal Daga>kubectl get hpa
No resources found in default namespace.

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myd-56d496f844-j6ldp               1/1     Running   0           7m26s
myd-56d496f844-q4n5w               1/1     Running   0           5m7s
myd-56d496f844-tttmr               1/1     Running   0           8m50s
myd-56d496f844-wqnx               1/1     Running   0           7m26s

C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
aws-node-52kft                     1/1     Running   0           44m
aws-node-962sq                     1/1     Running   0           44m
coredns-74ccfbd98b-cdj1l           1/1     Running   0           52m
coredns-74ccfbd98b-cgnrj           1/1     Running   0           52m
kube-proxy-4rcrg                   1/1     Running   0           44m
kube-proxy-ppqzc                   1/1     Running   0           44m

C:\Users\Vimal Daga>
```

Google

<https://github.com/kubernetes-sigs/metrics-server>

Installing the Kubernetes Metrics Server

The **Kubernetes Metrics Server** is an aggregator of resource usage data in your cluster, and it is not deployed by default in Amazon EKS clusters.

[GitHub](https://github.com)
<https://github.com/kubernetes-sigs/metrics-server>

[Kubernetes Metrics Server](#)

Metrics Server can be installed either directly from YAML manifest or via the official Helm chart. To install the latest Metrics Server release from the ...

[Releases 34](#) · [Issues 25](#) · [FAQ.md](#) · [README.md](#)

[GitHub Pages](#)
<https://kubernetes-sigs.github.io/metrics-server/>

[Kubernetes Metrics Server](#)

Kubernetes Metrics Server - A single deployment that works on most clusters (see Requirements) · Fast autoscaling, collecting metrics every 15 seconds. · Resource ...

[SigNoz](#)
<https://signoz.io/blog/kubernetes-metrics-server/>

[How to deploy k8s metrics server and use it for monitoring](#)

12-Aug-2022 — Kubernetes metrics-server is available on Github. Let's open that link and head over to the **Installation** section.

— [Medium](#)

Metrics Server can be installed either directly from YAML manifest or via the official [Helm chart](#). To install the latest Metrics Server release from the `components.yaml` manifest, run the following command.

```
kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
```

Installation instructions for previous releases can be found in [Metrics Server releases](#).

Compatibility Matrix

Metrics Server	Metrics API group/version	Supported Kubernetes version
0.6.x	metrics.k8s.io/v1beta1	1.19+
0.5.x	metrics.k8s.io/v1beta1	*1.8+

```
C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
aws-node-52kft                      1/1     Running   0           44m
aws-node-962sq                      1/1     Running   0           44m
coredns-74ccfbd98b-cdj1l            1/1     Running   0           52m
coredns-74ccfbd98b-cgnrj            1/1     Running   0           52m
kube-proxy-4rcrg                    1/1     Running   0           44m
kube-proxy-ppqzc                    1/1     Running   0           44m

C:\Users\Vimal Daga>kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
aws-node-52kft                      1/1     Running   0           46m
aws-node-962sq                      1/1     Running   0           46m
coredns-74ccfbd98b-cdj1l            1/1     Running   0           55m
coredns-74ccfbd98b-cgnrj            1/1     Running   0           55m
kube-proxy-4rcrg                    1/1     Running   0           46m
kube-proxy-ppqzc                    1/1     Running   0           46m
metrics-server-6bf466fbf5-gk9xr      0/1     Running   0           15s

C:\Users\Vimal Daga>kubectl get deployment -n kube-system
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
coredns         2/2     2             2           55m
metrics-server  1/1     1             1           33s

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubectl get deployment -n kube-system
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
coredns         2/2     2             2           55m
metrics-server  1/1     1             1           33s

C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
aws-node-52kft                      1/1     Running   0           46m
aws-node-962sq                      1/1     Running   0           46m
coredns-74ccfbd98b-cdj1l            1/1     Running   0           55m
coredns-74ccfbd98b-cgnrj            1/1     Running   0           55m
kube-proxy-4rcrg                    1/1     Running   0           46m
kube-proxy-ppqzc                    1/1     Running   0           46m
metrics-server-6bf466fbf5-gk9xr      1/1     Running   0           41s

C:\Users\Vimal Daga>
```

```

C:\Users\Vimal Daga>kubectl get deployment
NAME READY UP-TO-DATE AVAILABLE AGE
myd 4/4 4 4 13m

C:\Users\Vimal Daga>kubectl scale deployment myd --replicas=1
deployment.apps/myd scaled

C:\Users\Vimal Daga>kubectl get deployment
NAME READY UP-TO-DATE AVAILABLE AGE
myd 1/1 1 1 13m

C:\Users\Vimal Daga>

```

```

C:\Users\Vimal Daga>kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created

C:\Users\Vimal Daga>kubectl create deployment myd --image=vimal13/apache-webserver-php
deployment.apps/myd created

C:\Users\Vimal Daga>

```

```

C:\Users\Vimal Daga>kubectl get hpa
No resources found in default namespace.

C:\Users\Vimal Daga>kubectl autoscale deployment myd --max=10 --min=2 --cpu-percent=60
horizontalpodautoscaler.autoscaling/myd autoscaled

C:\Users\Vimal Daga>kubectl get hpa
NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
myd Deployment/myd <unknown>/60% 2 10 0 6s

C:\Users\Vimal Daga>kubectl get pods
NAME READY STATUS RESTARTS AGE
myd-7d544fd8bc-bcmn5 1/1 Running 0 47s

C:\Users\Vimal Daga>

```

```

C:\Users\Vimal Daga>kubectl get pods
NAME READY STATUS RESTARTS AGE
myd-7d544fd8bc-bcmn5 1/1 Running 0 106s
myd-7d544fd8bc-vljcm 1/1 Running 0 58s

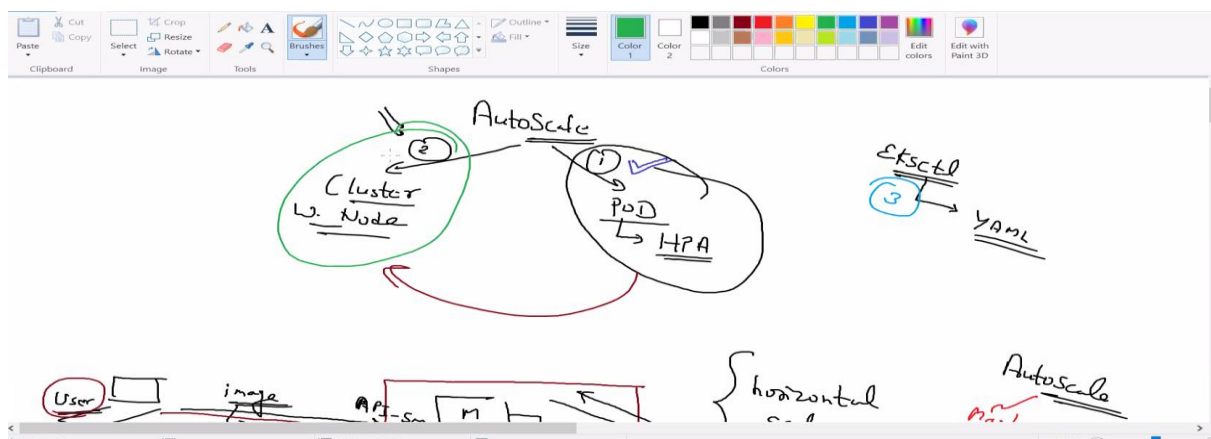
C:\Users\Vimal Daga>kubectl get hpa
NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
myd Deployment/myd <unknown>/60% 2 10 2 87s

C:\Users\Vimal Daga>

```

- Autoscaling of pod is not managed by EKS and AWS internally manage k8s.

- We as human beings have to monitor manually again and again so we don't want manually so for that we use autoscale of pod.
- K8s have one resource that is called HPA.
- HPA is a horizontal scaling of pod. HPA keeps on monitoring CPU and as soon as load increase it will scale out and as soon as load decrease, we scale in. For this minimum Requirement is to keep on checking metrics.
- We need metrics information because we are scaling. We need certain tool that keeps on retrieving metrics information and provide to HPA and for that we have to write rule for HPA and for that we have to install metrics server and that program will run on the pods.
- The metrics server will start monitoring the metrics of pod.
- HPA don't belong to EKS, it belongs to Kubernetes. We can also do HPA with yaml also.



```
C:\Users\Vimal Daga>eksctl get cluster
```

NAME	REGION	EKSCTL CREATED
cluster-asg	ap-south-1	True
cluster1	ap-south-1	True
cluster2	ap-south-1	True
exciting-wardrobe-1679738433	ap-south-1	True

```
C:\Users\Vimal Daga>kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
myd-7d544fd8bc-bcmn5	1/1	Running	0	10m
myd-7d544fd8bc-vljcm	1/1	Running	0	10m

```
C:\Users\Vimal Daga>kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
ip-192-168-12-233.ec2.internal	Ready	<none>	10m	v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal	Ready	<none>	10m	v1.25.7-eks-a59e1f0
ip-192-168-8-102.ec2.internal	Ready	<none>	10m	v1.25.7-eks-a59e1f0

```
C:\Users\Vimal Daga>kubectl describe nodes ip-192-168-12-233.ec2.internal
```

```

Name:         ip-192-168-12-233.ec2.internal
Roles:        <none>
Labels:       alpha.eksctl.io/cluster-name=my-cluster-asg
              alpha.eksctl.io/nodegroup-name=ngA
              beta.kubernetes.io/arch=amd64
              beta.kubernetes.io/instance-type=t2.micro
              beta.kubernetes.io/os=linux
              eks.amazonaws.com/capacityType=ON_DEMAND
              eks.amazonaws.com/nodegroup=ngA
              eks.amazonaws.com/nodegroup-image=ami-0a3da8b47de1d87b8
              eks.amazonaws.com/sourceLaunchTemplateId=lt-0ff3efdae3bfff6994
              eks.amazonaws.com/sourceLaunchTemplateVersion=1
              failure-domain.beta.kubernetes.io/region=us-east-1
              failure-domain.beta.kubernetes.io/zone=us-east-1b
              k8s.io/cloud-provider-aws=7261faf3ffb1a64c69a47ce05d78523e

```

Handwritten notes: $\frac{CPU}{T_{max}}$ and $\frac{2GB}{1GB}$

AWS

```
Select Command Prompt

C:\Users\Vimal Daga>eksctl get nodegroup --cluster my-cluster-asg --region us-east-1
CLUSTER      NODEGROUP    STATUS  CREATED              MIN SIZE  MAX SIZE  DESIRED CAPACITY
INSTANCE TYPE  IMAGE ID      ASG NAME
my-cluster-asg  ngA           ACTIVE  2023-04-02T10:55:48Z  2         10
t2.micro        AL2_x86_64    eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b managed

C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS  ROLES  AGE  VERSION
ip-192-168-12-233.ec2.internal      Ready  <none>  13m  v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal      Ready  <none>  13m  v1.25.7-eks-a59e1f0
ip-192-168-8-102.ec2.internal       Ready  <none>  13m  v1.25.7-eks-a59e1f0

C:\Users\Vimal Daga>
```

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:instanceState=running

Instances (3) Info

Find instance by attribute or tag (case-sensitive)

Instance state = running X Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
my-cluster-asg-ngA-Node	i-005d83ca040946d4a	Running	t2.micro	2/2 checks passed	No alarms
my-cluster-asg-ngA-Node	i-068b83bd150e09391	Running	t2.micro	2/2 checks passed	No alarms
my-cluster-asg-ngA-Node	i-0b219ef304f524912	Running	t2.micro	2/2 checks passed	No alarms

Select an instance

Auto Scaling groups (1/1) Info

Create an Auto Scaling group

Search your Auto Scaling groups

Name	Launch template/configuration	Instances	Status
eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	eks-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	3	-

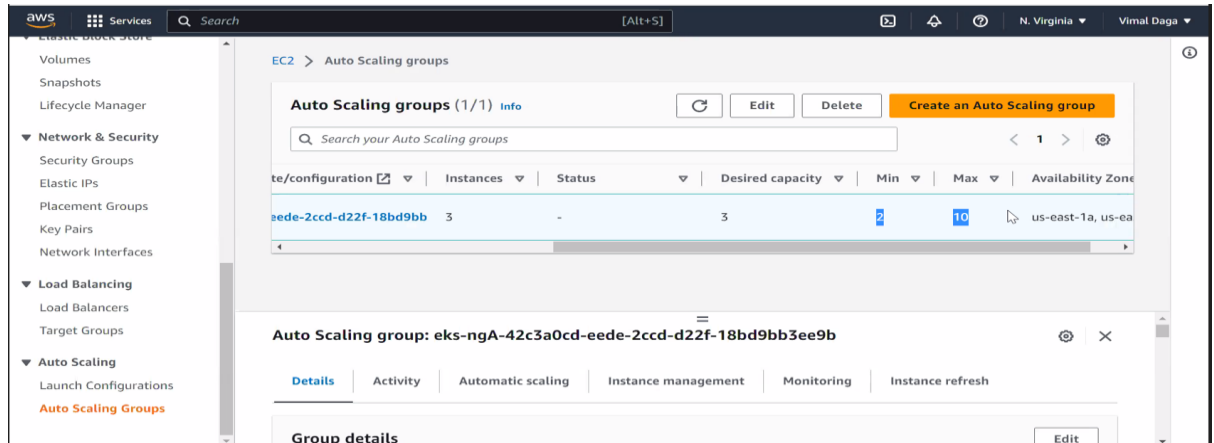
Auto Scaling group: eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b

Details Activity Automatic scaling Instance management Monitoring Instance refresh

Group details

Auto Scaling group name	Desired capacity	Status	Amazon Resource Name (ARN)
eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	3	-	arn:aws:autoscaling:us-east-1:583457386517:autoScaling

AWS



Auto Scaling groups (1/1) info

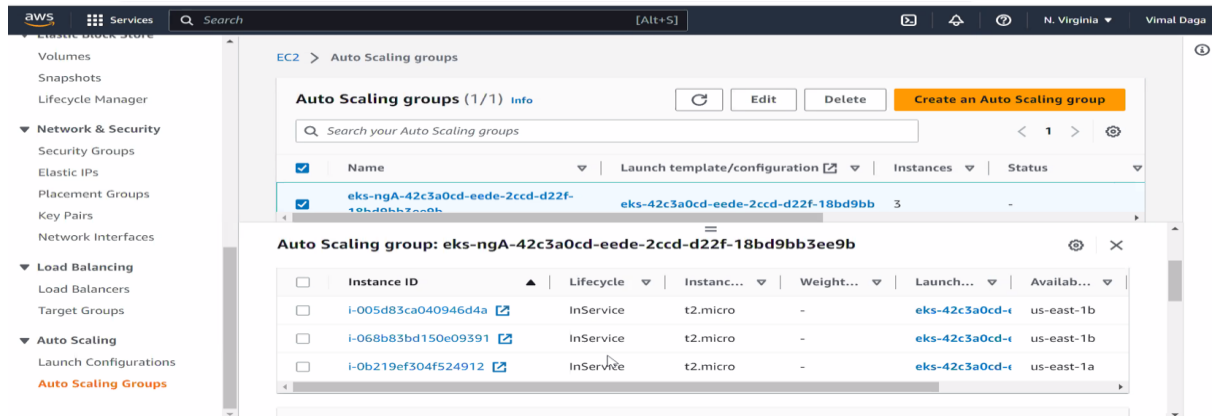
Search your Auto Scaling groups

Name/Configuration	Instances	Status	Desired capacity	Min	Max	Availability Zone
eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	3	-	3	2	10	us-east-1a, us-east-1b

Auto Scaling group: eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b

Details | Activity | Automatic scaling | Instance management | Monitoring | Instance refresh

Group details



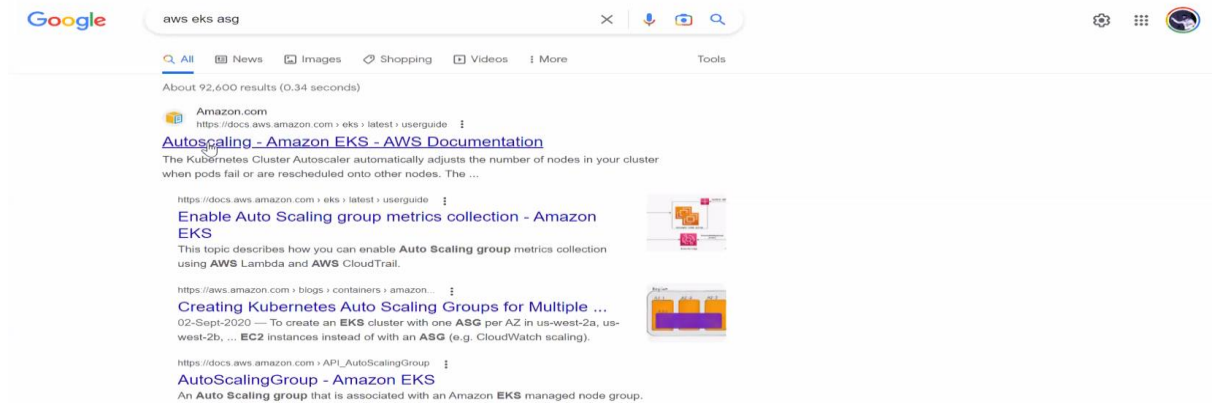
Auto Scaling groups (1/1) info

Search your Auto Scaling groups

Name	Launch template/Configuration	Instances	Status
eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	eks-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	3	-

Auto Scaling group: eks-ngA-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b

Instance ID	Lifecycle	Instance type	Weight	Launch template	Availability Zone
i-005d83ca040946d4a	InService	t2.micro	-	eks-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	us-east-1b
i-068b83bd150e09391	InService	t2.micro	-	eks-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	us-east-1b
i-0b219ef304f524912	InService	t2.micro	-	eks-42c3a0cd-eede-2ccd-d22f-18bd9bb3ee9b	us-east-1a



Google

aws eks asg

About 92,600 results (0.34 seconds)

Autoscaling - Amazon EKS - AWS Documentation

The Kubernetes Cluster Autoscaler automatically adjusts the number of nodes in your cluster when pods fail or are rescheduled onto other nodes. The ...

Enable Auto Scaling group metrics collection - Amazon EKS

This topic describes how you can enable Auto Scaling group metrics collection using AWS Lambda and AWS CloudTrail.

Creating Kubernetes Auto Scaling Groups for Multiple ...

02-Sept-2020 — To create an EKS cluster with one ASG per AZ in us-west-2a, us-west-2b, ... EC2 instances instead of with an ASG (e.g. CloudWatch scaling).

AutoScalingGroup - Amazon EKS

An Auto Scaling group that is associated with an Amazon EKS managed node group.

aws Search in this guide Contact Us English Return to the Console

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Autoscaling

- Nodes
- Storage
- Networking
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- Cluster authentication
- Cluster management
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 - View Kubernetes resources
- Observability
- Working with other services
- Troubleshooting
- Amazon EKS Connector
- Amazon EKS on AWS Outposts
- Related projects
- Amazon EKS new features and roadmap

Autoscaling

PDF RSS

Autoscaling is a function that automatically scales your resources up or down to meet changing demands. This is a major Kubernetes function that would otherwise require extensive human resources to perform manually.

Amazon EKS supports two autoscaling products. The Kubernetes [Cluster Autoscaler](#) and the [Karpenter](#) open source autoscaling project. The cluster autoscaler uses AWS scaling groups, while Karpenter works directly with the Amazon EC2 fleet.

Cluster Autoscaler

The Kubernetes [Cluster Autoscaler](#) automatically adjusts the number of nodes in your cluster when pods fail or are rescheduled

On this page

- [Cluster Autoscaler](#)
- Karpenter

```
C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-12-233.ec2.internal      Ready    <none>    17m    v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal      Ready    <none>    17m    v1.25.7-eks-a59e1f0
ip-192-168-8-102.ec2.internal       Ready    <none>    17m    v1.25.7-eks-a59e1f0

C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-12-233.ec2.internal      Ready    <none>    21m    v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal      Ready    <none>    21m    v1.25.7-eks-a59e1f0
ip-192-168-8-102.ec2.internal       Ready    <none>    21m    v1.25.7-eks-a59e1f0

C:\Users\Vimal Daga>curl -O https://raw.githubusercontent.com/kubernetes/autoscaler/master/cluster-autoscaler/cloudp
provider/aws/examples/cluster-autoscaler-autodiscover.yaml
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 4963  100 4963    0     0  9841    0 --:--:-- --:--:-- --:--:-- 9866

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>notepad cluster-autoscaler-autodiscover.yaml
```

```
apiVersion: v1
kind: ServiceAccount
metadata:
  labels:
    k8s-addon: cluster-autoscaler.addons.k8s.io
    k8s-app: cluster-autoscaler
  name: cluster-autoscaler
  namespace: kube-system

---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: cluster-autoscaler
  labels:
    k8s-addon: cluster-autoscaler.addons.k8s.io
    k8s-app: cluster-autoscaler
rules:
```

AWS

```
C:\Users\Vimal Daga>eksctl get cluster --region us-east-1
NAME          REGION    EKSCTL CREATED
my-cluster-asg us-east-1  True

C:\Users\Vimal Daga>
```

docs.aws.amazon.com/eks

```
nd Prompt
168-58-110.ec2.internal
168-8-102.ec2.internal

s\Vimal Daga>kubectl get
168-12-233.ec2.internal
168-58-110.ec2.internal
168-8-102.ec2.internal

s\Vimal Daga>curl -O http
/aws/examples/cluster-aut
al % Received % Xferd
53 100 4963 0 0

s\Vimal Daga>notepad clus
s\Vimal Daga>eksctl get c
REGION
ter-asg us-east-1

s\Vimal Daga>
on EKS Connector
on EKS on AWS Outposts
id projects
on EKS new features and
tap
```

caler/enabled,k8s.io/cluster-autoscaler/my-cluster-asg

certs/ca-bundle.crt for Amazon Linux Worker Nodes

Ln 165, Col 125 100% Unix (LF) UTF-8

```
Command Prompt
C:\Users\Vimal Daga>kubectl apply -f cluster-autoscaler-autodiscover.yaml
serviceaccount/cluster-autoscaler created
clusterrole.rbac.authorization.k8s.io/cluster-autoscaler created
role.rbac.authorization.k8s.io/cluster-autoscaler created
clusterrolebinding.rbac.authorization.k8s.io/cluster-autoscaler created
rolebinding.rbac.authorization.k8s.io/cluster-autoscaler created
deployment.apps/cluster-autoscaler created

C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY    STATUS    RESTARTS   AGE
aws-node-pbzs2                      1/1      Running   0           29m
aws-node-q7qq1                      1/1      Running   0           29m
aws-node-v22h1                      1/1      Running   0           29m
cluster-autoscaler-84d7bcf94b-5fx6r 0/1      Pending   0           15s
coredns-7975d6fb9b-csnjv            1/1      Running   0           39m
coredns-7975d6fb9b-tmkgj            1/1      Running   0           39m
kube-proxy-5wcc8                    1/1      Running   0           29m
kube-proxy-ff8gn                    1/1      Running   0           29m
kube-proxy-pptfz                    1/1      Running   0           29m
metrics-server-55c774cddb-pqcrp     1/1      Running   0           26m

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY    STATUS    RESTARTS   AGE
aws-node-pbzs2                      1/1      Running   0           30m
aws-node-q7qq1                      1/1      Running   0           30m
aws-node-v22h1                      1/1      Running   0           30m
cluster-autoscaler-84d7bcf94b-5fx6r 0/1      Pending   0           80s
coredns-7975d6fb9b-csnjv            1/1      Running   0           40m
coredns-7975d6fb9b-tmkgj            1/1      Running   0           40m
kube-proxy-5wcc8                    1/1      Running   0           30m
kube-proxy-ff8gn                    1/1      Running   0           30m
kube-proxy-pptfz                    1/1      Running   0           30m
metrics-server-55c774cddb-pqcrp     1/1      Running   0           27m

C:\Users\Vimal Daga>kubectl describe pods cluster-autoscaler-84d7bcf94b-5fx6r -n kube-system
```


AWS

```
Command Prompt
Volumes:
ssl-certs:
  Type: HostPath (bare host directory volume)
  Path: /etc/ssl/certs/ca-bundle.crt
  HostPathType:
kube-api-access-6hrp1:
  Type: Projected (a volume that contains injected data from multiple sources)
  TokenExpirationSeconds: 3607
  ConfigMapName: kube-root-ca.crt
  ConfigMapOptional: <nil>
  DownwardAPI: true
QoS Class: Guaranteed
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute for 300s
              node.kubernetes.io/unreachable:NoExecute for 300s
Events:
  Type      Reason      Age   From      Message
  ----      -
Warning    FailedScheduling 103s  default-scheduler 0/3 nodes are available: 2 Too many pods, 3 Insufficient memory.
preemption: 0/3 nodes are available: 1 No preemption victims found for incoming pod, 2 Insufficient memory.

C:\Users\Vimal Daga>
```

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ConnectToInstance:instanceId=i-005d83ca040946d4a

Services Search [Alt+S] N. Virginia Vimal Daga

New EC2 Experience

Instances (1/3) Info

Find instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	my-cluster-asg-ngA-Node	i-005d83ca040946d4a	Running	t2.micro	2/2 checks passed	No alarms
☐	my-cluster-asg-ngA-Node	i-068b83bd150e09391	Running	t2.micro	2/2 checks passed	No alarms
☐	my-cluster-asg-ngA-Node	i-0b219ef304f524912	Running	t2.micro	2/2 checks passed	No alarms

Instance: i-005d83ca040946d4a (my-cluster-asg-ngA-Node)

Details Security Networking Storage Status checks Monitoring Tags

Instance summary Info

Instance ID i-005d83ca040946d4a (my-cluster-asg-ngA-Node)	Public IPv4 address 35.172.200.208 open address	Private IPv4 addresses 192.168.26.102 192.168.12.233
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-35-172-200-208.compute-

```
aws
Services Search [Alt+S] N. Virginia Vimal Daga

Last login: Wed Mar 22 01:17:00 2023 from 205.251.233.106

 _ _ _ _ _
| | | | |
|_|_|_|_|_|_ Amazon Linux 2 AMI

https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-192-168-12-233 ~]$ free -m
              total        used        free      shared  buff/cache   available
Mem:           952         238          74           1         639         573
Swap:            0           0           0

[ec2-user@ip-192-168-12-233 ~]$ echo 3 > /proc/sys/vm/drop_caches
-bash: /proc/sys/vm/drop_caches: Permission denied
[ec2-user@ip-192-168-12-233 ~]$ sudo su -
[root@ip-192-168-12-233 ~]# cluster-autoscaler-84d7bcf94b-5fx6r^C
[root@ip-192-168-12-233 ~]# echo 3 > /proc/sys/vm/drop_caches
[root@ip-192-168-12-233 ~]# free -m
              total        used        free      shared  buff/cache   available
Mem:           952         226          479           1         247         590
Swap:            0           0           0

[root@ip-192-168-12-233 ~]#
```

```

aws
Services Search
Last login: Wed Mar 22 01:17:00 20
Amazon Linux
https://aws.amazon.com/amazon-linu
[ec2-user@ip-192-168-12-233 ~]$ fr
total used
Mem: 952 238
Swap: 0 0
[ec2-user@ip-192-168-12-233 ~]$ ec
-bash: /proc/sys/vm/drop_caches: P
[ec2-user@ip-192-168-12-233 ~]$ su
[root@ip-192-168-12-233 ~]# cluste
[root@ip-192-168-12-233 ~]# echo 3
[root@ip-192-168-12-233 ~]# free -
total used
Mem: 952 226
Swap: 0 0
[root@ip-192-168-12-233 ~]#

runAsNonRoot: true
runAsUser: 65534
fsGroup: 65534
seccompProfile:
  type: RuntimeDefault
serviceAccountName: cluster-autoscaler
containers:
- image: registry.k8s.io/autoscaling/cluster-autosc
  name: cluster-autoscaler
resources:
  limits:
    cpu: 100m
    memory: 600Mi
  requests:
    cpu: 100m
    memory: 200Mi
command:
- ./cluster-autoscaler

```

```

Command Prompt
DownwardAPI: true
QoS Class: Guaranteed
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute for 300s
              node.kubernetes.io/unreachable:NoExecute for 300s
Events:
  Type      Reason              Age   From          Message
  ----      -
Warning    FailedScheduling    103s  default-scheduler  0/3 nodes are available: 2 Too many pods, 3 Insufficient memory.
y. preemption: 0/3 nodes are available: 1 No preemption victims found for incoming pod, 2 Insufficient memory.

C:\Users\Vimal Daga>notepad cluster-autoscaler-autodiscover.yaml

C:\Users\Vimal Daga>kubectl apply -f cluster-autoscaler-autodiscover.yaml
serviceaccount/cluster-autoscaler unchanged
clusterrole.rbac.authorization.k8s.io/cluster-autoscaler unchanged
role.rbac.authorization.k8s.io/cluster-autoscaler unchanged
clusterrolebinding.rbac.authorization.k8s.io/cluster-autoscaler unchanged
rolebinding.rbac.authorization.k8s.io/cluster-autoscaler unchanged
deployment.apps/cluster-autoscaler configured

C:\Users\Vimal Daga>

```

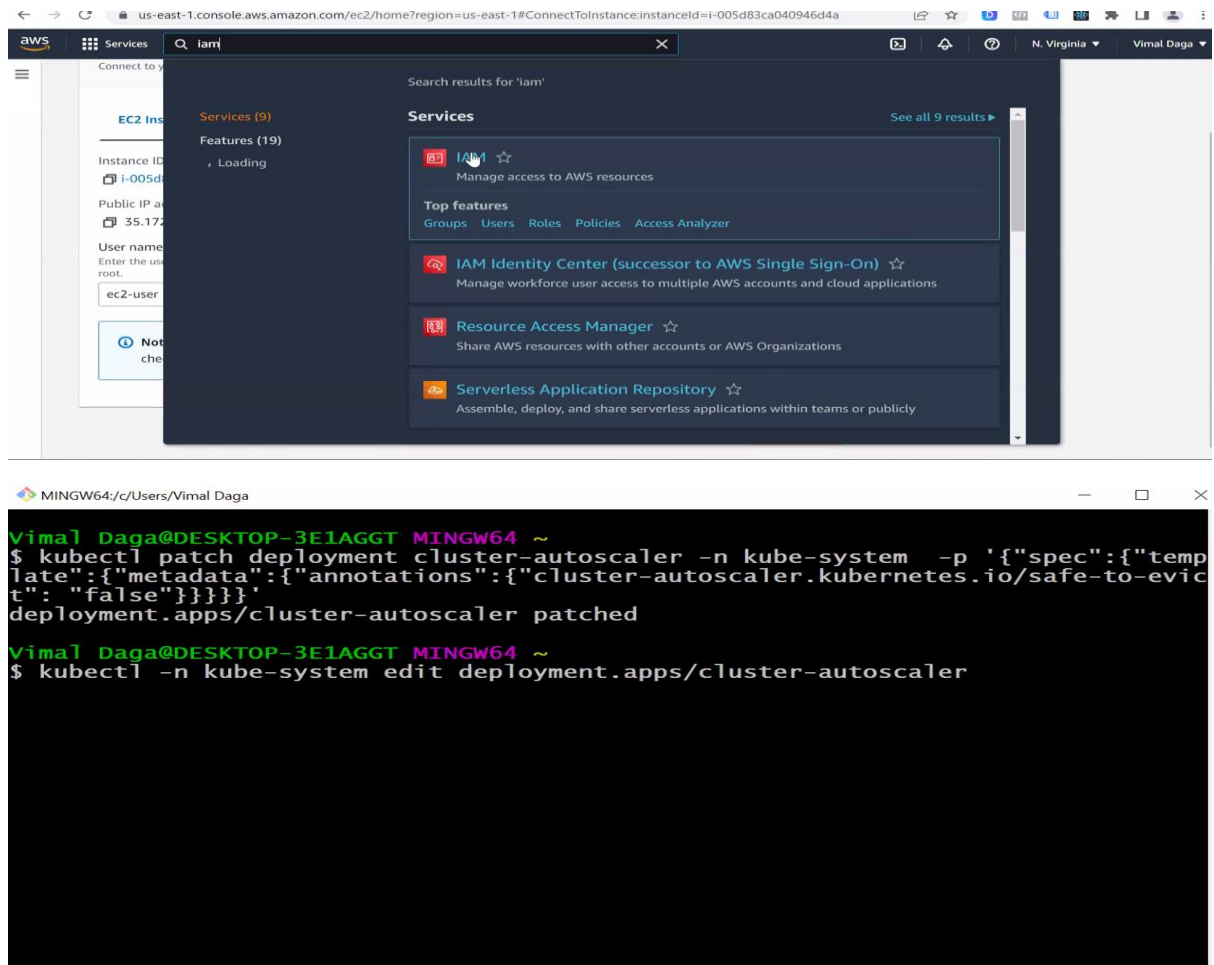
```

C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY    STATUS    RESTARTS   AGE
aws-node-pbzsx                      1/1      Running   0           35m
aws-node-q7qq1                      1/1      Running   0           35m
aws-node-v22h1                      1/1      Running   0           35m
cluster-autoscaler-fd4f4f55c-szmk4  1/1      Running   0           12s
coredns-7975d6fb9b-csnjv            1/1      Running   0           45m
coredns-7975d6fb9b-tmkgj            1/1      Running   0           45m
kube-proxy-5wcc8                    1/1      Running   0           35m
kube-proxy-ff8gn                    1/1      Running   0           35m
kube-proxy-pptfz                    1/1      Running   0           35m
metrics-server-55c774cd8b-pqcrp     1/1      Running   0           32m

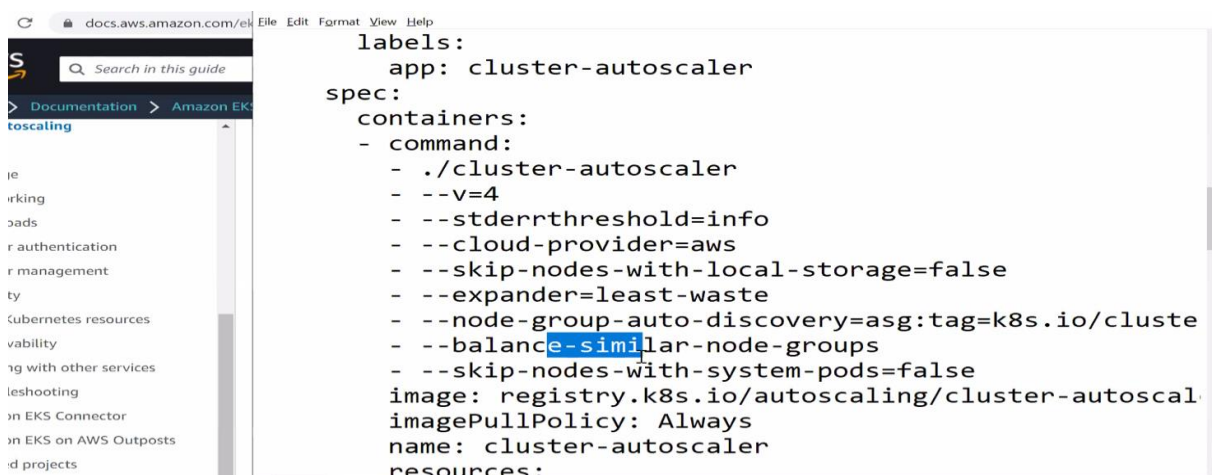
C:\Users\Vimal Daga>

```

AWS



- Cluster autoscaler keeps on monitoring and sends information to ASG. It captures entire information and that information sent to the ASG and based upon the information it keeps on launching and terminating instance.
- There are two ways to autoscale
 1. Karpenter
 2. ASG(Default)




```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~
$ kubectl set image deployment cluster-autoscaler -n kube-system cluster-autoscaler=registry.k8s.io/autoscaling/cluster-autoscaler:v1.25.1
deployment.apps/cluster-autoscaler image updated
```

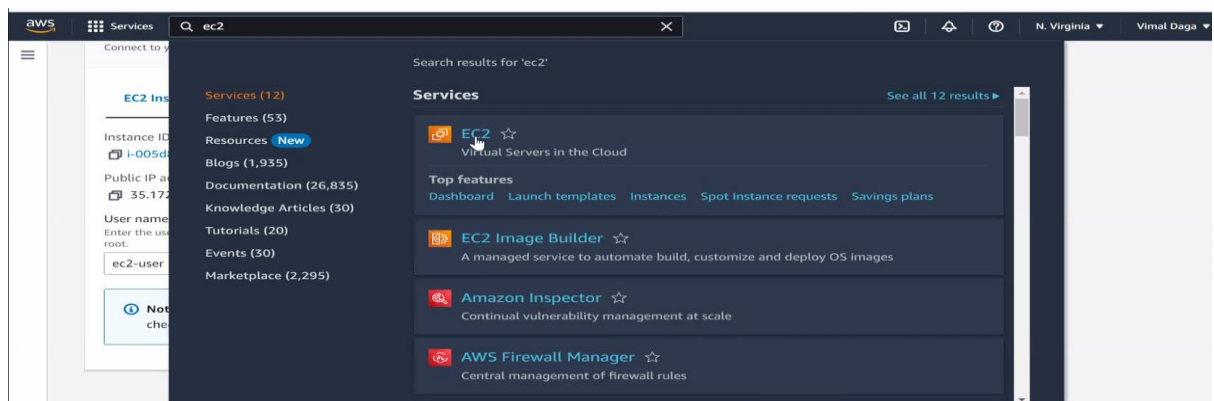
```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~
$
```

```
C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
aws-node-f5cr6                      1/1     Running   0           4m28s
aws-node-pbzsx                      1/1     Running   0           46m
aws-node-q7qq1                      1/1     Running   0           46m
aws-node-v22h1                      1/1     Running   0           46m
cluster-autoscaler-6bc889fb5c-9fzqx 1/1     Running   0           12s
coredns-7975d6fb9b-csnjv            1/1     Running   0           56m
coredns-7975d6fb9b-tmkgj            1/1     Running   0           56m
kube-proxy-5wcc8                    1/1     Running   0           46m
kube-proxy-ff8gn                    1/1     Running   0           46m
kube-proxy-h76hz                    1/1     Running   0           4m28s
kube-proxy-pptfz                    1/1     Running   0           46m
metrics-server-55c774cddb-pqcrp     1/1     Running   0           43m

C:\Users\Vimal Daga>kubectl -n kube-system logs -f deployment.apps/cluster-autoscaler
```

```
Command Prompt - kubectl -n kube-system logs -f deployment.apps/cluster-autoscaler
I0402 11:45:02.115674    1 cluster.go:160] ip-192-168-8-102.ec2.internal for removal
I0402 11:45:02.115757    1 cluster.go:251] Pod kube-system/coredns-7975d6fb9b-tmkgj can be moved to ip-192-168-12-233.ec2.internal
I0402 11:45:02.115863    1 cluster.go:251] Pod kube-system/coredns-7975d6fb9b-csnjv can be moved to ip-192-168-52-72.ec2.internal
I0402 11:45:02.115882    1 cluster.go:182] node ip-192-168-8-102.ec2.internal may be removed
I0402 11:45:02.115909    1 legacy.go:510] ip-192-168-58-110.ec2.internal is unneeded since 2023-04-02 11:44:52.014080648 +0000 UTC m=+31.711076846 duration 10.100590534s
I0402 11:45:02.115943    1 legacy.go:510] ip-192-168-12-233.ec2.internal is unneeded since 2023-04-02 11:44:52.014080648 +0000 UTC m=+31.711076846 duration 10.100590534s
I0402 11:45:02.115956    1 legacy.go:510] ip-192-168-8-102.ec2.internal is unneeded since 2023-04-02 11:44:52.014080648 +0000 UTC m=+31.711076846 duration 10.100590534s
I0402 11:45:02.115979    1 static_autoscaler.go:535] Scale down status: lastScaleUpTime=2023-04-02 10:44:42.01297505 +0000 UTC m=-3578.290028689 lastScaleDownDeleteTime=2023-04-02 10:44:42.01297505 +0000 UTC m=-3578.290028689 scaleDownForbidden=false scaleDownInCooldown=false
I0402 11:45:02.116001    1 static_autoscaler.go:544] Starting scale down
I0402 11:45:02.116027    1 legacy.go:628] ip-192-168-58-110.ec2.internal was unneeded for 10.100590534s
I0402 11:45:02.116061    1 legacy.go:628] ip-192-168-12-233.ec2.internal was unneeded for 10.100590534s
I0402 11:45:02.116072    1 legacy.go:628] ip-192-168-8-102.ec2.internal was unneeded for 10.100590534s
I0402 11:45:02.116087    1 legacy.go:727] No candidates for scale down
```

```
C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-12-233.ec2.internal      Ready    <none>    48m    v1.25.7-eks-a59e1f0
ip-192-168-52-72.ec2.internal       Ready    <none>    6m33s  v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal      Ready    <none>    48m    v1.25.7-eks-a59e1f0
ip-192-168-8-102.ec2.internal       Ready    <none>    48m    v1.25.7-eks-a59e1f0
```



AWS

Instances (4) Info

Find instance by attribute or tag (case-sensitive)

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
my-cluster-asg-ngA-Node	i-005d83ca040946d4a	Running	t2.micro	2/2 checks passed	No alarms
my-cluster-asg-ngA-Node	i-068b83bd150e09391	Running	t2.micro	2/2 checks passed	No alarms
my-cluster-asg-ngA-Node	i-0b219ef304f524912	Running	t2.micro	2/2 checks passed	No alarms
my-cluster-asg-ngA-Node	i-081a2f0beb33e3f75	Running	t2.micro	2/2 checks passed	No alarms

Select an instance

Instances (4) Info

Find instance by attribute or tag (case-sensitive)

Instance state = running

IP	IPv6 IPs	Monitoring	Security group name	Key name	Launch time
-	-	disabled	eksctl-my-cluster-asg-n...	eksctl-my-clus...	2023/04/02 16:27 GMT+5:30
-	-	disabled	eksctl-my-cluster-asg-n...	eksctl-my-clus...	2023/04/02 16:27 GMT+5:30
-	-	disabled	eksctl-my-cluster-asg-n...	eksctl-my-clus...	2023/04/02 16:27 GMT+5:30
-	-	disabled	eksctl-my-cluster-asg-n...	eksctl-my-clus...	2023/04/02 17:09 GMT+5:30

```
C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY    STATUS    RESTARTS   AGE
aws-node-f5cr6                      1/1     Running   0           9m43s
aws-node-pbzs2                      1/1     Running   0           51m
aws-node-q7qq1                      1/1     Running   0           51m
aws-node-v22h1                      1/1     Running   0           51m
cluster-autoscaler-6bc889fb5c-9fzqx 1/1     Running   0           5m27s
coredns-7975d6fb9b-csnjv           1/1     Running   0           61m
coredns-7975d6fb9b-tmkgj           1/1     Running   0           61m
kube-proxy-5wcc8                    1/1     Running   0           51m
kube-proxy-ff8gn                    1/1     Running   0           51m
kube-proxy-h76hz                    1/1     Running   0           9m43s
kube-proxy-pptfz                    1/1     Running   0           51m
metrics-server-55c774cddb-pqcrp     1/1     Running   0           48m
```

```
C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-12-233.ec2.internal      Ready    <none>    52m    v1.25.7-eks-a59e1f0
ip-192-168-52-72.ec2.internal       Ready    <none>    11m    v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal      Ready    <none>    53m    v1.25.7-eks-a59e1f0
ip-192-168-8-102.ec2.internal       Ready    <none>    53m    v1.25.7-eks-a59e1f0

C:\Users\Vimal Daga>eksctl get cluster
NAME                                REGION    EKSTCL CREATED
cluster-asg                         ap-south-1    True
cluster1                            ap-south-1    True
cluster2                            ap-south-1    True
exciting-wardrobe-1679738433       ap-south-1    True

C:\Users\Vimal Daga>eksctl get cluster --region us-east-1
NAME                                REGION    EKSTCL CREATED
my-cluster-asg                      us-east-1    True

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubect1 get hpa
NAME          REFERENCE          TARGETS          MINPODS          MAXPODS          REPLICAS          AGE
myd           Deployment/myd      <unknown>/60%   2                10              2                52m

C:\Users\Vimal Daga>kubect1 delete all --all
pod "myd-7d544fd8bc-bcmn5" deleted
pod "myd-7d544fd8bc-kvt4f" deleted
service "kubernetes" deleted
deployment.apps "myd" deleted
horizontalpodautoscaler.autoscaling "myd" deleted

C:\Users\Vimal Daga>kubect1 get hpa
No resources found in default namespace.

C:\Users\Vimal Daga>kubect1 get pods
No resources found in default namespace.

C:\Users\Vimal Daga>notepad php.yml
```

AWS

```
Microsoft Windows [Version 10.0.19044.2728]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Vimal Daga>kubectl get hpa php-apache --watch
NAME      REFERENCE                TARGETS  MINPODS  MAXPODS  REPLICAS  AGE
php-apache Deployment/php-apache     0%/50%   1         10        1          3m43s
php-apache Deployment/php-apache     22%/50%   1         10        1          4m16s
```

```
Microsoft Windows [Version 10.0.19044.2728]
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C:\Users\Vimal Daga>kubectl get pods
NAME                READY   STATUS    REPLICAS
load-generator      1/1     Running   1
php-apache-5b56f9df94-b5xnn 0/1     Pending   0
php-apache-5b56f9df94-pxzmp 1/1     Running   1
php-apache-5b56f9df94-s7z56 0/1     Pending   0
php-apache-5b56f9df94-xzmfz 0/1     Pending   0

C:\Users\Vimal Daga>kubectl get hpa php-apache --watch
NAME      REFERENCE                TARGETS  MINPODS  MAXPODS  REPLICAS  AGE
php-apache Deployment/php-apache     0%/50%   1         10        1          3m43s
php-apache Deployment/php-apache     22%/50%   1         10        1          4m16s
php-apache Deployment/php-apache     249%/50%  1         10        1          4m31s
```

```
Microsoft Windows [Version 10.0.19044.2728]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Vimal Daga>kubectl get pods
NAME                READY   STATUS    REPLICAS
load-generator      1/1     Running   1
php-apache-5b56f9df94-b5xnn 0/1     Pending   0
php-apache-5b56f9df94-pxzmp 1/1     Running   1
php-apache-5b56f9df94-s7z56 0/1     Pending   0
php-apache-5b56f9df94-xzmfz 0/1     Pending   0
php-apache-5b56f9df94-zv6w8 0/1     Pending   0

C:\Users\Vimal Daga>kubectl get pods
NAME                READY   STATUS    REPLICAS
load-generator      1/1     Running   1
php-apache-5b56f9df94-b5xnn 0/1     Pending   0
php-apache-5b56f9df94-pxzmp 1/1     Running   1
php-apache-5b56f9df94-s7z56 0/1     Pending   0
php-apache-5b56f9df94-xzmfz 0/1     Pending   0
php-apache-5b56f9df94-zv6w8 0/1     Pending   0

C:\Users\Vimal Daga>kubectl get hpa php-apache --watch
NAME      REFERENCE                TARGETS  MINPODS  MAXPODS  REPLICAS  AGE
php-apache Deployment/php-apache     0%/50%   1         10        1          3m43s
php-apache Deployment/php-apache     22%/50%   1         10        1          4m16s
php-apache Deployment/php-apache     249%/50%   1         10        1          4m31s
php-apache Deployment/php-apache     250%/50%   1         10        4          4m46s
php-apache Deployment/php-apache     249%/50%   1         10        5          5m1s
```

```
php-apache-5b56f9df94-xzmfz 0/1     Pending
php-apache-5b56f9df94-zv6w8 0/1     Pending

C:\Users\Vimal Daga>kubectl get hpa php-apache --watch
NAME      REFERENCE                TARGETS  MINPODS  MAXPODS  REPLICAS  AGE
php-apache Deployment/php-apache     250%/50%  1         10        5          5m55s
```

```
C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-12-233.ec2.internal      Ready    <none>    67m   v1.25.7-eks-a59e1f0
ip-192-168-24-39.ec2.internal       Ready    <none>    67s   v1.25.7-eks-a59e1f0
ip-192-168-52-72.ec2.internal       Ready    <none>    25m   v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal      Ready    <none>    67m   v1.25.7-eks-a59e1f0
ip-192-168-7-33.ec2.internal        Ready    <none>    69s   v1.25.7-eks-a59e1f0

C:\Users\Vimal Daga>kubectl get pods
NAME      REFERENCE                TARGETS  MINPODS  MAXPODS  REPLICAS
php-apache Deployment/php-apache     250%/50%  1         10        5
```



```

C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE      VERSION
ip-192-168-12-233.ec2.internal      Ready    <none>   70m      v1.25.7-eks-a59e1f0
ip-192-168-24-39.ec2.internal       Ready    <none>   4m19s    v1.25.7-eks-a59e1f0
ip-192-168-37-80.ec2.internal       Ready    <none>   109s     v1.25.7-eks-a59e1f0
ip-192-168-52-72.ec2.internal       Ready    <none>   28m      v1.25.7-eks-a59e1f0
ip-192-168-58-110.ec2.internal      Ready    <none>   70m      v1.25.7-eks-a59e1f0
ip-192-168-7-33.ec2.internal        Ready    <none>   4m21s    v1.25.7-eks-a59e1f0

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
load-generator                      1/1      Running   0           6m17s
php-apache-5b56f9df94-4bvbt        1/1      Running   0           3m16s
php-apache-5b56f9df94-b5xnn        1/1      Running   0           5m46s
php-apache-5b56f9df94-pxzmp        1/1      Running   0           10m
php-apache-5b56f9df94-s7z56        1/1      Running   0           5m46s
php-apache-5b56f9df94-v244d        1/1      Running   0           2m46s
php-apache-5b56f9df94-xzmfz        1/1      Running   0           5m46s
php-apache-5b56f9df94-zv6w8        1/1      Running   0           5m31s

C:\Users\Vimal Daga>

```

```

C:\Users\Vimal Daga>kubectl get hpa php-apache --watch
NAME                                REFERENCE    TARGETS    MINPODS    MAXPODS    REPLICAS    AGE
php-apache                         Deployment/php-apache  47%/50%    1          10         7           10m

```

```

C:\Users\Vimal Daga>kubectl get pods -n kube-system
NAME                                READY    STATUS    RESTARTS   AGE
aws-node-5m8tb                      1/1      Running   0           6m37s
aws-node-f5cr6                      1/1      Running   0           30m
aws-node-grng7                      1/1      Running   0           6m39s
aws-node-pbzs                          1/1      Running   0           72m
aws-node-v22h1                      1/1      Running   0           72m
aws-node-xmr8p                      1/1      Running   0           4m7s
cluster-autoscaler-6bc889f15c-9fzqx 1/1      Running   0           26m
coredns-7975d6fb9b-xx1s6           1/1      Running   0           16m
coredns-7975d6fb9b-zb22s           1/1      Running   0           16m
kube-proxy-5wcc8                    1/1      Running   0           72m
kube-proxy-cvs2x                    1/1      Running   0           6m37s
kube-proxy-h76hz                    1/1      Running   0           30m
kube-proxy-hkwcc                    1/1      Running   0           4m7s
kube-proxy-mtb4r                    1/1      Running   0           6m39s
kube-proxy-pptfz                    1/1      Running   0           72m
metrics-server-55c774cddb-pqcrp     1/1      Running   0           69m

```

- So by this scaling of pod will be done automatically.
- If load increase, replica of pod will also increase with the help of hpa and if more resources are required it will autoscale the pod.